

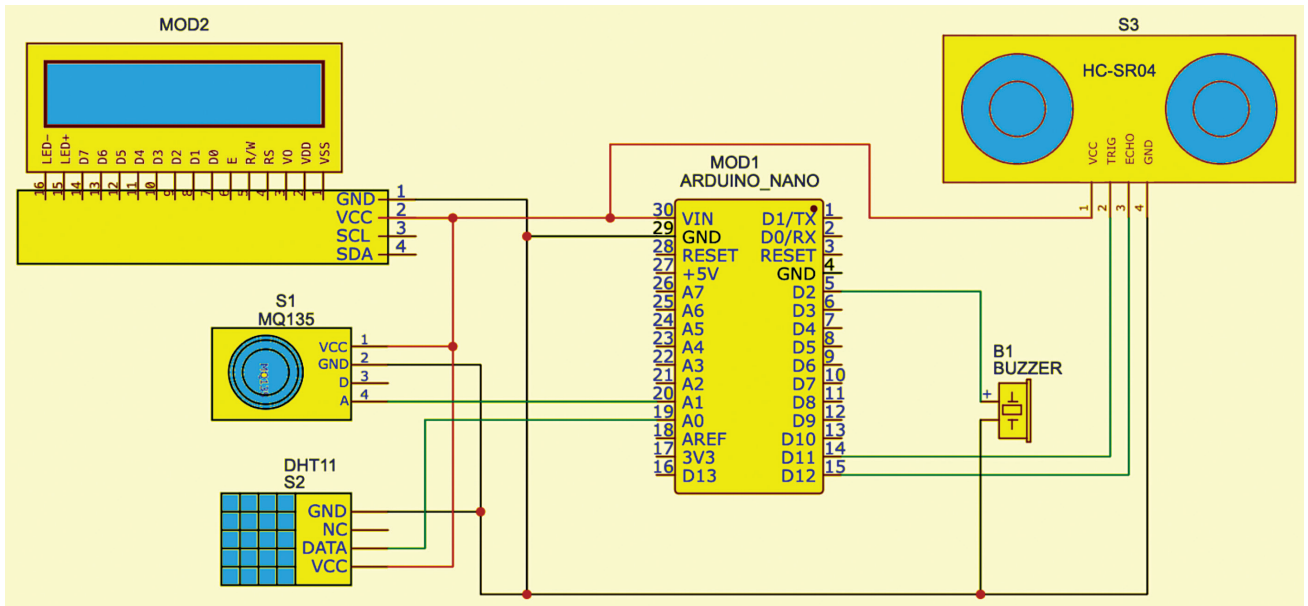


Fig. 2: Circuit diagram

```

sketch_sep16a.ino
1  #include <dht.h>
2  #include <LiquidCrystal_I2C.h>
3  LiquidCrystal_I2C lcd(0x27,16,2);
4  dht DHT;
5  #define DHT11_PIN A0
6  #define sensor A1
7  #define trigPin 11
8  #define echoPin 12
9  void setup(){
10   lcd.init();
11   lcd.clear();
12   lcd.backlight(); // Make sure backlight is on
13   lcd.setCursor(3,0);
14   lcd.print("SMART");
15   lcd.setCursor(4,1);
16   lcd.print("EcoBin");
17   pinMode(sensor,INPUT);
18   pinMode(2,OUTPUT);
19   pinMode(trigPin, OUTPUT);
20   pinMode(echoPin, INPUT);
21   delay(2000);
22 }
23
24 void loop(){
25   long duration, distance;
26   digitalWrite(trigPin, LOW);
27   delayMicroseconds(2);
28   digitalWrite(trigPin, HIGH);
29   delayMicroseconds(10);
30   digitalWrite(trigPin, LOW);
31   duration = pulseIn(echoPin, HIGH);
32   distance = (duration/2) / 29.1;
33   if (distance < 5) {
34     delay(1000);
35     digitalWrite(2, HIGH);
36     lcd.setCursor(3,0);
37     lcd.print("EcoBin is");
38     lcd.setCursor(3,1);
39     lcd.print("Full");
40   }
41 }

```

Fig. 3: Code snippet

ideal for breadboard-based systems such as the Smart EcoBin. The Nano can be programmed using the Arduino IDE and connected to a computer via a Mini USB cable for

code uploading and serial monitoring.

For coding, the required libraries DHT.h (for the DHT11 sensor) and LiquidCrystal_I2C.h (for the I2C LCD module) must be installed using the Arduino IDE Library Manager. These libraries are then included in the code, and the pins are defined according to circuit connections. In this setup, A0 is used for the DHT11 sensor, A1 for the MQ135 gas sensor, pins 11 and 12 for the ultrasonic distance sensor, and pin 2 for the buzzer. These assignments can be modified, but the code must be updated accordingly to reflect the changes. For the I2C LCD, A4 (SDA) and A5 (SCL) are fixed communication pins and must be used.

After setting up the pins, the code checks sensor data, processes readings, and displays results on the LCD. Functions are created to trigger alerts, such as sounding the buzzer when the EcoBin is full, when poor air

quality is detected, or if temperature rise indicates a potential fire. By structuring the program in modular functions, the Smart EcoBin remains easy to manage, debug, and extend. Fig. 3 shows a code snippet.

Construction and testing

The process begins by uploading the source code into the Arduino Nano using the Arduino IDE, ensuring that the correct board and COM port are selected. The Nano is placed at the centre of the breadboard for convenient connections. As shown in Fig. 2, the I2C LCD module is wired by connecting its SDA and SCL pins to A4 and A5, respectively, with power provided through the 5V and GND lines. The MQ135 gas sensor is connected to its analogue output on A1, and the DHT11 sensor is connected to its data pin on A0. The prototype, after connection to the breadboard, is shown in Fig. 4.

The HC-SR04 ultrasonic sensor is mounted on one side of the breadboard, linking TRIG and ECHO to D11 and D12, respectively. The buzzer is attached to D2, with its second terminal grounded. All modules share the common 5V and GND rails from the Nano to ensure